Directions for Questions 1 to 4: *Consider the following graph (Figure 4.1) and answer the questions based on it. The graph shows the trend of consumption of metals and plastics in the production of cars between 2007 and 2012.*

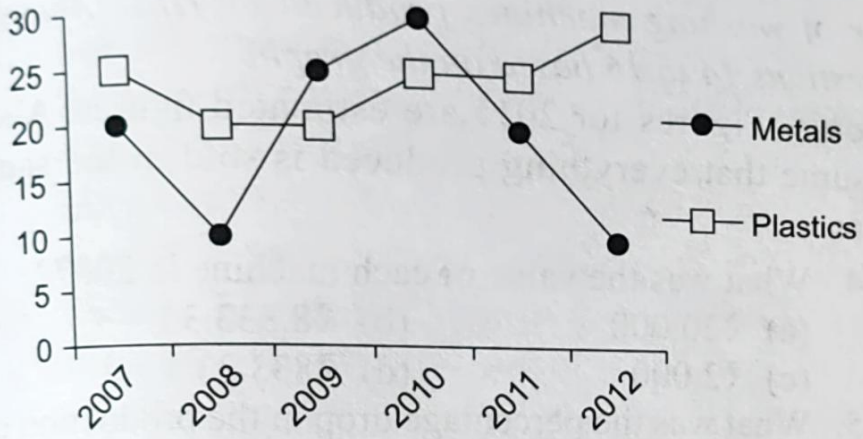


FIGURE 4.1 *Consumption of Metals versus Plastics in the given Years for Car Manufacturing (in thousand tonnes)*

1. The total consumption of plastic (for car manufacturing) divided by the total consumption of metals (for car manufacturing) over the period will give a ratio the closest to:
   (a) 4:3
   (b) 5:4
   (c) 6:5
   (d) 7:4
2. The ratio of the highest total consumption in any single year to the lowest total consumption of the two items taken together in any year was equal to:
   (a) 5:3
   (b) 7:4
   (c) 11:6
   (d) 11:7
3. Which year showed the highest percentage increase in the total consumption of the two metals?
   (a) 2008
   (b) 2009
   (c) 2010
   (d) 2011

4. Which year showed the highest percentage decrease in the total consumption of the 2 items taken together over the previous year?
   (a) 2008
   (b) 2009
   (c) 2011
   (d) 2012

Directions for Questions 22 to 23: *Study the following graph (Figure 4.7) carefully and answer the given questions. Assume that the curve between any two years marked on the graph is a straight line and the difference between any two years can be got by interpolation. Beyond 2003 the values go up to 2005 and are the projected values for 2004 and 2005.*

22. In 1963, the percentage of HBS students who were engineers was about
    (a) 18% (b) 25%
    (c) 30% (d) 35%
23. Which was the first year when more than 200 students were studying in a batch at HBS?
    (a) 1973 (b) 1978
    (c) 1983 (d) 1985

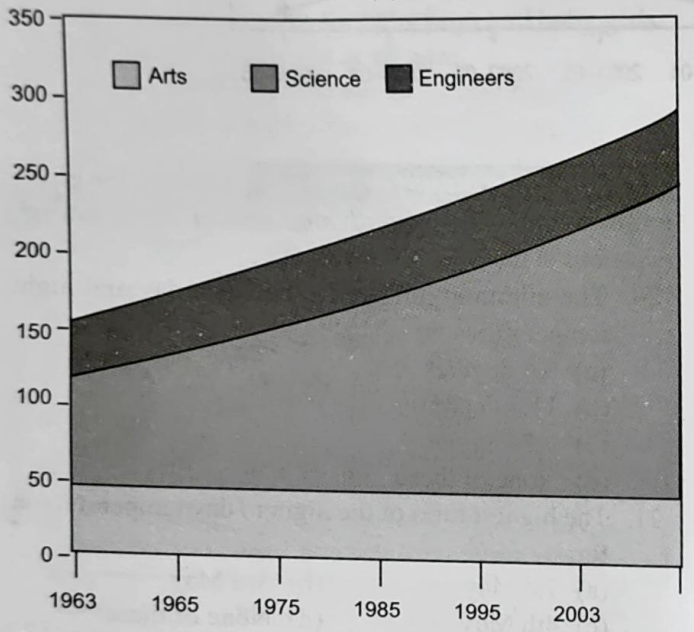


FIGURE 4.7 *Academic Background of Students Joining Harvard Business School (HBS)*

Directions for Question 24: *The following graph (Figure 4.8) gives the movement of a car with respect to time. (Assume that the speed dimension on the Y-axis is marked at every 10 kmph and there is uniform acceleration between hours.)*

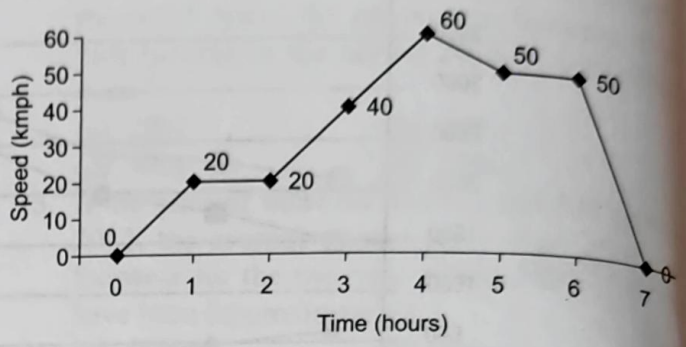


FIGURE 4.8 *Movement of a Car with Respect to Time*

24. Find the average speed of the car (in kmph).
    (a) 36.28
    (b) 34.28
    (c) 33.57
    (d) None of these

Directions for Questions 25 to 28: *These questions are based on the situation given below:*

The figure below presents sales and net profit, in ₹ crore, of IVP Ltd. for the five years from 1994–95 to 1998–99. During this period, the sales increased from ₹100 crore to ₹680 crore. Correspondingly, the net profit increased from ₹2.2 crores to ₹12 crore. Net profit is defined as the excess of sales over total costs.

Note: Figure 4.9 has two Y-axis – one on the left representing the sales and the one on the right representing profit. The Y-axis on the right has a scale of 1 unit representing ₹2 crore.

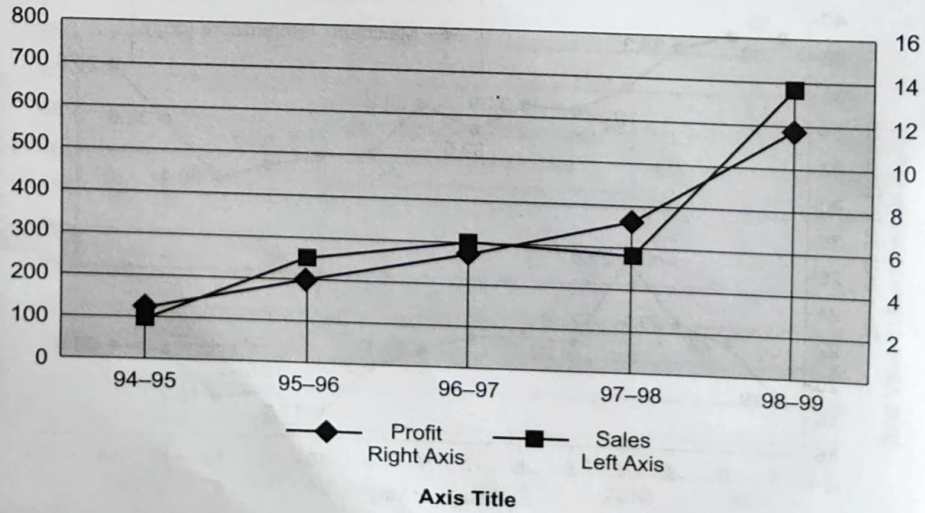


FIGURE 4.9

25. The highest percentage of growth in sales, relative to the previous year, occurred in:
    (a) 1995–96 (b) 1996–97
    (c) 1997–98 (d) Can't be determined
26. The highest percentage growth in net profit, relative to the previous year, was achieved in:
    (a) 1998–99 (b) 1997–98
    (c) 1996–97 (d) 1995–96
27. Defining profitability as the ratio of net profit to sales, IVP Ltd. recorded the highest profitability in
    (a) 1998–99 (b) 1997–98
    (c) 1994–95 (d) 1996–97